

Letter of Recommendation for Neva Krien

Date: May 18, 2026

From: Stephen Blum, Co-founder and CTO, PubNub

Re: Neva Krien, AI/ML, Data Science, and Software Engineer

To Whom It May Concern,

I am writing to recommend **Neva Krien** for an AI/ML, Data Science, or Software Engineering role on your team. I am Stephen Blum, CTO and co-founder of PubNub. I first met Neva in 2025 during my live YouTube tech presentations, where I cover programming language topics and walk through code on stream. Across multiple presentations since then, Neva has consistently contributed in real time at an expert level, and in that time I have come to know her as one of the sharpest engineers I have had the chance to learn alongside.

Neva's guidance came in at the right moments. She understands what is essential in a system and what is noise. We read through her code on stream, and it held up cleanly.

Neva has continued to contribute in steady, meaningful ways. She wrote a PyTorch guide that I worked through on a follow-up stream, and when I was preparing to dig into a topic she could not be available for, she opened a PR with the answer ready so the live video stream would not stall. Most recently, she guided the audience, explaining **Embedding layers**, one of those concepts that sounds simple until you actually try to build intuition for what the vectors are doing and why. Neva walked us through it in a way that finally clicked.

Hands-On Model Building Together

From more code reviews and guides, Neva and I have built models together using the PyTorch framework. This is where I got the clearest signal on her engineering instincts.

We worked through **CNNs for image classification**, defining the conv/pool/activation stack, reasoning about kernel sizes and channel depth, and stepping through how the receptive field grows layer by layer. Neva was sharp on the tradeoffs: when to add batch norm, when dropout actually helps, and how to tell whether a training curve is telling you the truth or lying to you.

From there we moved into **embedding layers for NLP**, building text models using PyTorch embeddings end-to-end, tokenization, vocabulary construction, embedding lookup, and feeding the dense vectors into downstream layers. Neva explained why embeddings work the way they do, not just how to call `nn.Embedding`. She made the geometry of the vector space concrete, which is the part most tutorials skip.

We also built a **click prediction model** together, feature engineering for sparse categorical inputs, embedding the high-cardinality fields, combining them with dense numerical features, and training against a binary objective. Neva had strong intuitions about class imbalance, calibration, and what evaluation metrics actually tell you in a recommendation/ranking context versus what people often default to.

Across the experiences shared, what stood out is that Neva does not just stitch together PyTorch APIs. She knows what each layer is doing mathematically, why the architecture is shaped the way it is, and where it will break under real-world data. That is the difference between someone who has done a tutorial and someone who can ship ML in production.

What Stands Out

A few things stand out about working with her. She writes clean, readable code, in a field where it is easy to drown in framework abstractions, Neva produces examples that strip a problem down to its core, which is exactly the skill you want in someone building or maintaining ML systems. She teaches well: explaining ML and computer science concepts to someone live, on stream, with viewers watching, is a real test of how well you actually understand a topic, and Neva is consistently clear, patient, and accurate. She knows the difference between handwaving and explaining.

She is also proactive, she does not wait to be asked, she sees a gap, builds the thing, and shares it. That is the engineer I want on my team. And she is genuinely curious. Her work spans Rust, PyTorch, ML internals, and systems-level thinking. She is not chasing whatever is trending. She is going deep.

Recommendation

If you are hiring for an AI/ML, data science or software engineering role, Neva is the kind of person who will ship real work, lift the people around her, and ask the right questions about what you are actually trying to build. **I recommend her without hesitation.**

Please feel free to reach out if you would like to discuss further.

Sincerely,

Stephen Blum Co-founder and CTO, PubNub

<https://www.linkedin.com/in/stephenlb>

A handwritten signature in black ink, appearing to read 'Stephen Blum', with a stylized, cursive script.